Bilkent University
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Lab Report 03
EE102 - 01

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Verification of Combinational Logic Using a Synchronous 4-Bit Binary Counter (74HC163)

Abstract:
The objective of this lab session was to learn to verify the real-time operation of a digital combinational logic circuit driven by a synchronous binary counter. A 74HC163 4-bit counter was utilized to generate a sequential binary progression from a 1 Hz clock signal. The outputs were fed into a combinational logic network constructed using 74HC08 (AND) and 74HC32 (OR) logic gates to realize the Boolean function ; $F = (Y_0 \cdot Y_1) + Y_2$.

Oscilloscope measurements and an active-LOW visual LED indicator helped confirm timing analysis, demonstrating an output that remained LOW for three clock cycles and HIGH for five clock cycles, perfectly matching the derived truth table.

1. Introduction

Sequential logic elements, such as synchronous counters, are foundational in generating deterministic state progressions in digital systems. By interfacing these sequential outputs with combinational logic gates, specific Boolean functions can be evaluated in real-time. The objective of this laboratory exercise is to construct a hardware realization of a specific Boolean expression, verify the frequency division of the synchronous counter, and quantitatively analyze the timing of the final combinational output using a digital storage oscilloscope.

2. Theory

The circuit utilizes a 74HC163 synchronous 4-bit binary counter. The input clock frequency f_{clk} is divided by 2^n at each consecutive output stage Y_n. For our purposes of the experiment, only the first three stages are utilized:

- Y_0(Pin 14): Least Significant Bit (LSB),$f = f_{clk} / 2$
- Y_1 (Pin 13): $f = f_{clk} / 4$
- Y_2 (Pin 12): Most Significant Bit (MSB) for the target logic, $f = f_{clk} / 8$

The combinational logic circuit is designed to implement the following Boolean function:

$$F = (Y_0 \cdot Y_1) + Y_2$$

Because the circuit repeats its cycle every 8 clock pulses (2^3 states), the logic can be fully mapped across states 0 through 7.

Table 1: Truth Table for the Combinational Logic

Decimal State	**Y2 (Pin 12)**	**Y1 (Pin 13)**	**Y0 (Pin 14)**	**(Y0 · Y 1)**	**Output F**	**Active-LOW LED**
0	0	0	0	0	**0**	**ON**
1	0	0	1	0	**0**	**ON**
2	0	1	0	0	**0**	**ON**
3	0	1	1	1	**1**	OFF
4	1	0	0	0	**1**	OFF

5	1	0	1	0	**1**	OFF
6	1	1	0	0	**1**	OFF
7	1	1	1	1	**1**	OFF

Timing Prediction: Given a $1\ Hz$ input clock (period $T = 1s$), the output F is mathematically predicted to remain LOW for $3.0s$ (States 0, 1, 2) and HIGH for $5.0s$ (States 3, 4, 5, 6, 7).

3. Methodology

3.1. Equipments and Components

Sequential IC: 74HC163 (4-bit Synchronous Binary Counter).
Combinational ICs: 74HC08 (Quad 2-Input AND), 74HC32 (Quad 2-Input OR).
Measurement: Agilent InfiniiVision DSO-X 2002A Digital Storage Oscilloscope.
Signal Generation: $1\ Hz$ square wave, $5.00\ V_{pp}$, $2.5V$ offset.
Visual Output: Red LED connected in an active-LOW configuration with a 330Ω pull-up resistor.
Power Supply: DC Power Supply set to $5.0\ V$.

3.2. Circuit Schematic

The logic circuit was designed to interface the sequential counter outputs with a combinational logic gate network. The signal routing directs the 74HC163 Y_0 and Y_1 outputs through the 74HC08 AND gate. That product, along with Y_2, is fed into the 74HC32 OR gate to evaluate $F = (Y_0 \cdot Y_1) + Y_2$.

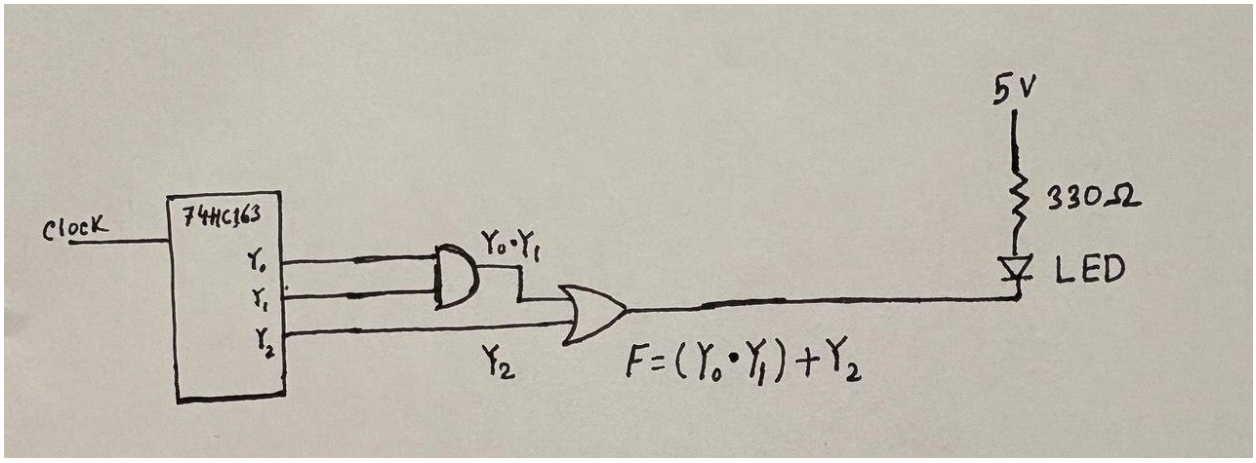

Figure 1: *Logic schematic diagram illustrating the 4-bit synchronous counter driving the combinational logic circuit. The final output terminates in an active-LOW LED configuration.*

3.3. Circuit Construction & Verification

The theoretical schematic was physically realized on a standard breadboard. The 74HC163 control pins ($\overline{MR}$, $\overline{PE}$, $CEP,\ CET$) were tied to the $5V\ \ V_{CC}$ rail to enable continuous synchronous running.

To provide real-time visual verification of the logic states, an active-LOW LED circuit was constructed at the final output. A 330 Ω resistor was connected directly from the $5V\ \ V_{CC}$rail to the anode of a red LED. The cathode of the LED was connected directly to the output of the 74HC32 OR gate. In this configuration, the logic gate acts as a current sink. When the OR gate outputs a logic HIGH ($5V$), the potential difference across the LED is zero, keeping it OFF. When the logic output transitions to a LOW state ($0V$), current flows from the V_{CC} rail through the LED, turning it ON.

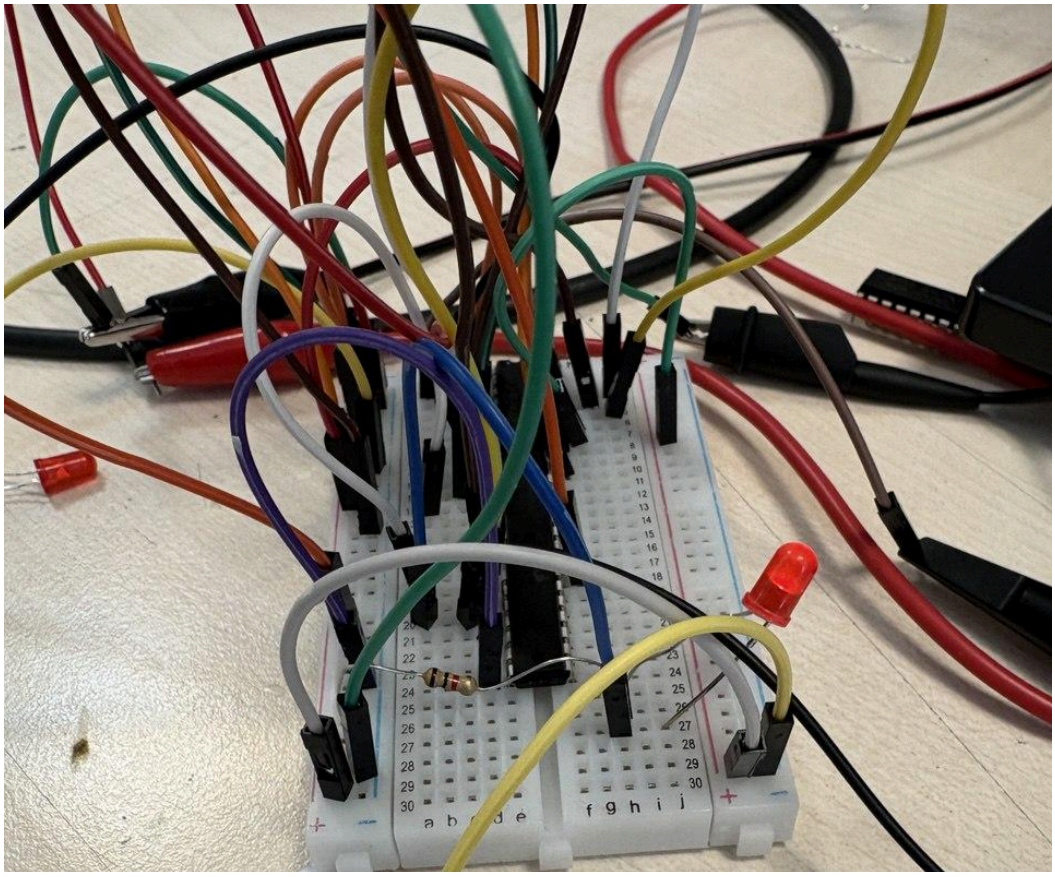

Figure 2: *Physical realization of the logic schematic on a breadboard. The setup includes the 74HC163 counter, 74HC08 AND gate, and 74HC32 OR gate, with jumper wires bridging the split power rails to ensure proper power distribution.*

4. Results and Analysis

4.1. Sequential Frequency Division

The outputs of the 74HC163 counter were probed independently to verify proper state transitions before implementing the combinational logic. The oscilloscope showed the following frequency halving:

Input Clock: $1000.00\ mHz$

Y_0 Output: $500.00\ mHz$

Y_1 Output: $250.00\ mHz$

Y_2 Output: $125.00\ mHz$

Y_3 Output: $62.50\ mHz$

The peak-to-peak voltage across all output stages were measured between $4.8\ V$ and $5.1\ V$. This confirms achievement of robust TTL-compatible high states.

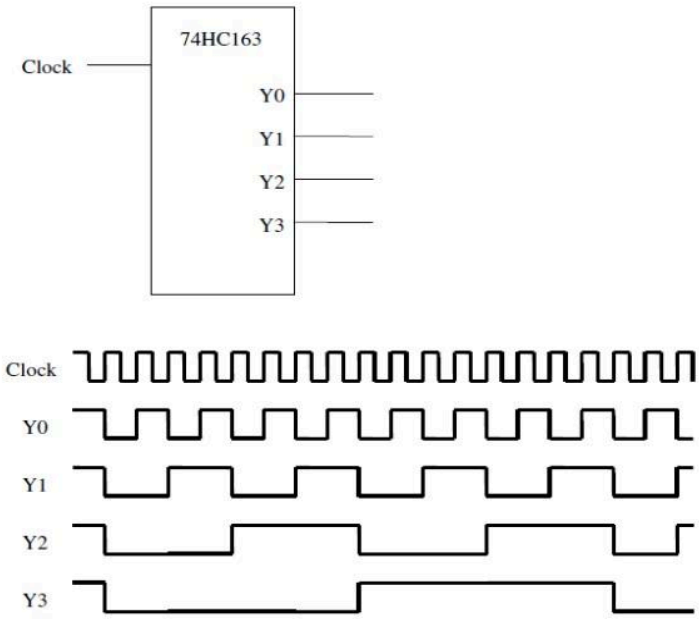

Figure 3: Expected waveforms for outputs of the 74HC163 counter

Figure 4: Starting from top-left, obtained waveforms of Y_3, Y_2, Y_1, Y_0 *and Clock from the 74HC163 counter. Oscilloscope captures demonstrate the accurate frequency division of the 74HC163 counter at the output pins prior to logic evaluation.*

4.2. Combinational Timing Verification

To verify the evaluated function F, CH1 (Yellow) was used to monitor the $1\ Hz$ input clock, while CH2 (Green) was used to monitor the output of the 74HC32 OR gate.

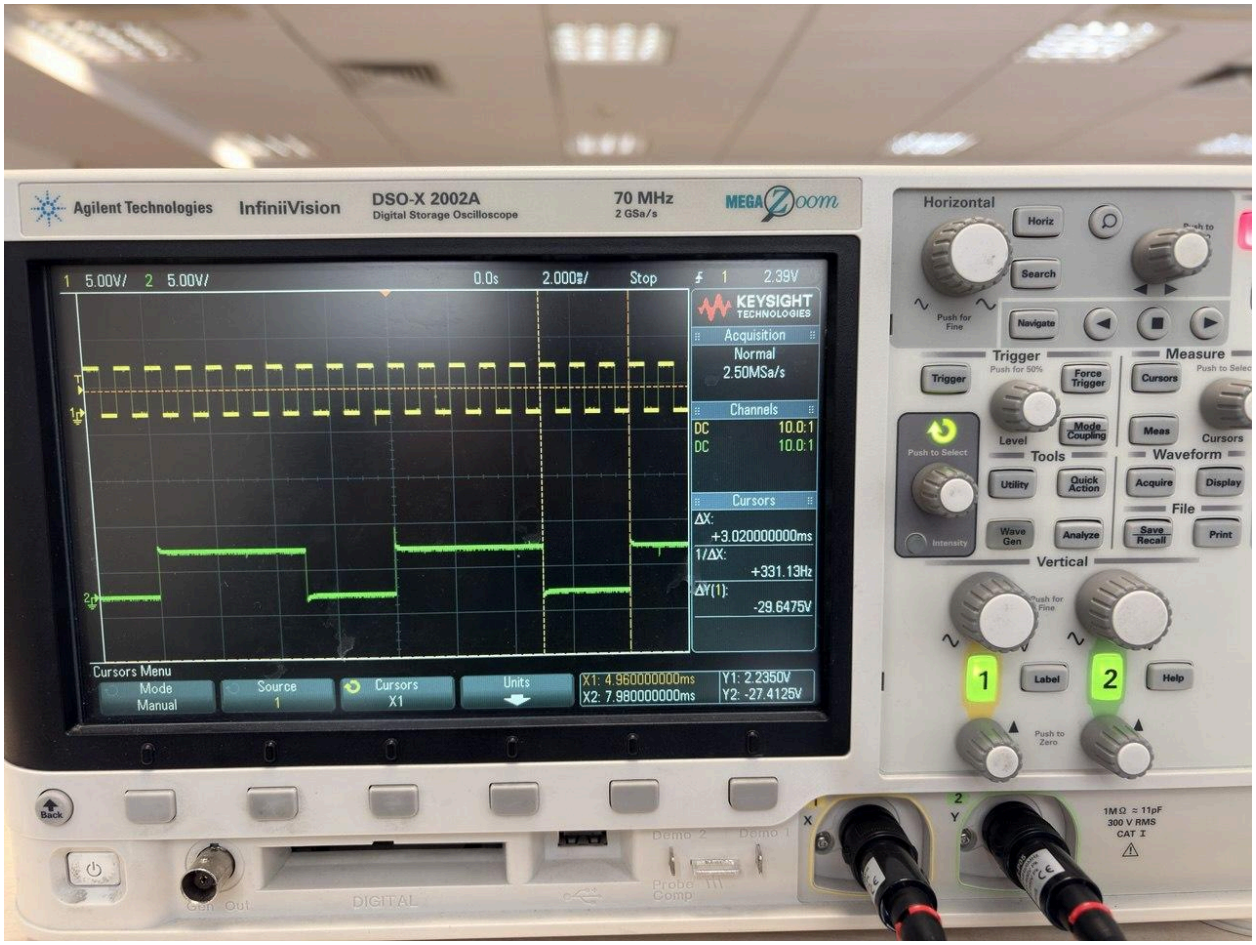

Figure 5: Verification of the final combinational logic output F. The oscilloscope cursors confirm the output remains LOW for three clock cycles and HIGH for the subsequent five cycles.

The resulting waveform aligned with the theoretical timing diagram. The oscilloscope's manual cursors were utilized to measure the duration of the logic LOW state.

Theoretical LOW Duration: $3.00\ s$
Measured LOW Duration (ΔX): $3.02\ s$
Error Percentage: 0.67%

The 0.67% error obtained is statistically negligible and is attributed to the internal tolerance of the laboratory function generator and manual cursor placement on the digital oscilloscope's screen.

5. Hardware Considerations

During the initial implementation phase, critical hardware parameters required debugging to align the physical circuit with the theoretical schematic:

1. Power Distribution Integrity: The ICs initially exhibited floating output states. This was due to a physical discontinuity in the breadboard's internal power and ground rails. Jumper wires (M-M) were installed to bridge the central gap, ensuring a common $5V$ and Ground reference across all ICs.
2. Logic Gate Completeness: The Boolean expression required both AND and OR operations. Initial tests failed due to attempting to pass all signals exclusively through the 74HC32 (OR) IC, resulting in logic contention. The integration of the dedicated 74HC08 (AND) IC resolved this architectural error.
3. Control Pin Biasing: The 74HC163 failed to increment from State 0 due to an improperly biased Count Enable (CEP) pin. In high-speed CMOS (HC) logic, floating or grounded enable pins act as active deterrents to clock synchronous functions. Tying CEPdirectly to $5V$ restored normal sequential operation.

6. Conclusion

This laboratory session obtained its objective in demonstrating the physical integration of sequential counting hardware with a combinational logic design. The theoretical Boolean expression $F = (Y_0 \cdot Y_1) + Y_2$ was accurately realized. Quantitative analysis using the digital storage oscilloscope gave a logic LOW pulse width of $3.02s$ against a theoretical expected value of $3.00s$. This, within slight instrumental error range, confirmed the precise accuracy of the logic circuit. The visual active-LOW LED indicator verified this operation in real-time by illuminating exactly during the predicted 0 logic states, thoroughly following the derived truth table for the combinational logic circuit.

References

1. Philips Semiconductors, "74HC/HCT163 Presettable synchronous 4-bit binary counter; synchronous reset," *Integrated Circuits Data Sheet*, IC06, Dec. 1990.

Appendix A: Integrated Circuit Pin Configurations

To provide context for the physical wiring described in the methodology section, the standard dual in-line package (DIP) pin configurations for the discrete logic ICs utilized in this experiment are detailed below.

A.1. 74HC163 (Presettable Synchronous 4-Bit Binary Counter)

According to the Philips Semiconductors specification:

Pin 1: $\overline{MR}$ (Synchronous Master Reset, Active LOW) – Tied to $5V$
Pin 2: CP (Clock Input, LOW-to-HIGH edge-triggered) – Connected to $1Hz$ Function Generator
Pins 3, 4, 5, 6: D_0, D_1, D_2, D_3(Data Inputs) – Tied to Ground
Pin 7:CEP (Count Enable Input) – Tied to $5V$
Pin 8: GND (Ground, $0V$)
Pin 9: $\overline{PE}$ (Parallel Enable Input, Active LOW) – Tied to $5V$
Pin 10: CET (Count Enable Carry Input) – Tied to $5V$
Pins 14, 13, 12, 11: Q_0, Q_1, Q_2, Q_3 (Flip-Flop Outputs Y_0, Y_1, Y_2, Y_3)
Pin 15: TC(Terminal Count Output) – Unused
Pin 16:V_{CC} (Positive Supply Voltage, $5V$)

A.2. 74HC08 (Quad 2-Input AND Gate)

Pin 1: $1A$ Input (Connected to Y_0)
Pin 2: $1B$ Input (Connected to Y_1)
Pin 3: $1Y$ Output ($Y_0 \cdot Y_1$)
Pin 7: GND (Ground)
Pin 14:V_{CC} ($5V$ Power)

A.3. 74HC32 (Quad 2-Input OR Gate)

Pin 1: $1A$ Input (Connected to AND Output $1Y$)
Pin 2: $1B$ Input (Connected to Y_2)
Pin 3: $1Y$ Final Logic Output (Connected to LED Cathode)
Pin 7: GND (Ground)
Pin 14: V_{CC} ($5V$ Power)